2024

The Implications of a Dying World



Wilson, Damian T

Cal State University Los Angeles

12/12/2024

Data Set URL:

https://www.kaggle.com/datasets/sevgisarac/temperature-change?resource=download

INTRODUCTION-

Global warming, primarily driven by human activities, is one of the most pressing challenges of the 21st century. It refers to the long-term increase in Earth's average surface temperature due to the buildup of greenhouse gases (GHGs) like carbon dioxide (CO_2), methane (CH_4), and nitrous oxide (N_2O) in the atmosphere. These gases trap heat, leading to rising temperatures, melting ice caps, and extreme weather patterns. As the planet warms, the consequences become more severe, including rising sea levels, loss of biodiversity, and disruptions to agriculture and water resources (Deman,2024). Addressing global warming requires comprehensive strategies that focus on both mitigation (reducing emissions) and adaptation (adjusting to its effects).

The most effective strategy for combating global warming is reducing the emissions of GHGs, which largely come from the burning of fossil fuels for energy. A major focus is on transitioning to renewable energy sources such as wind, solar, and hydropower, which produce little to no GHG emissions. According to the International Renewable Energy Agency (IRENA), renewable energy is central to decarbonizing the global economy. Additionally, enhancing energy efficiency across all sectors, transportation, buildings, and industry—can significantly lower energy consumption and emissions.

Carbon capture technologies aim to capture CO_2 emissions from industrial processes before they are released into the atmosphere and store them underground or in other long-term storage sites. This technology is seen as essential for decarbonizing sectors like cement and steel production, which are challenging to fully electrify. CCS is considered a vital component of climate strategies to achieve net-zero emissions.

Forests play a critical role in absorbing carbon from the atmosphere. Therefore, reforestation (restoring deforested areas) and afforestation (planting new forests) are important strategies for increasing carbon sequestration. Programs like the United Nations' REDD+ aim to reduce emissions from deforestation and promote sustainable land management practices. These efforts not only mitigate climate change but also enhance biodiversity and protect ecosystems.

Agriculture contributes significantly to global warming through methane emissions from livestock and rice paddies, and nitrous oxide from fertilizers. Adopting sustainable farming practices such as precision agriculture, crop rotation, and agroforestry can reduce

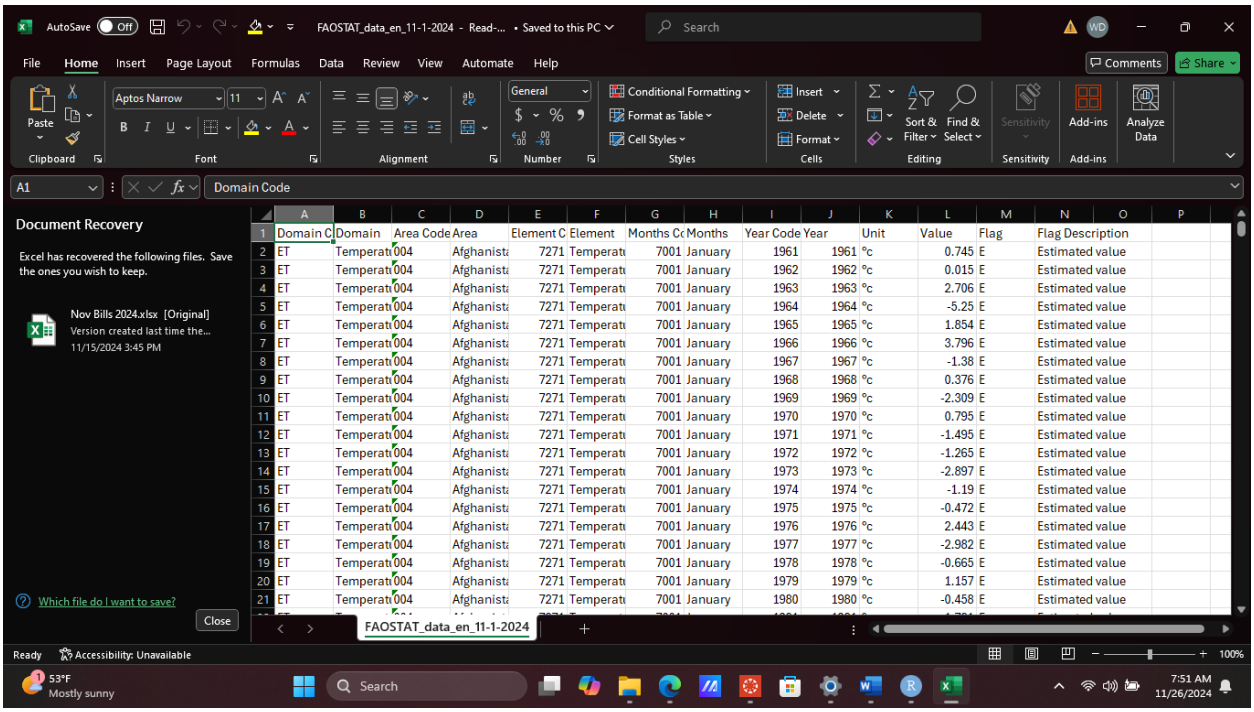
emissions while maintaining food security. Moreover, increasing carbon sequestration through techniques like no-till farming helps capture carbon in the ground.

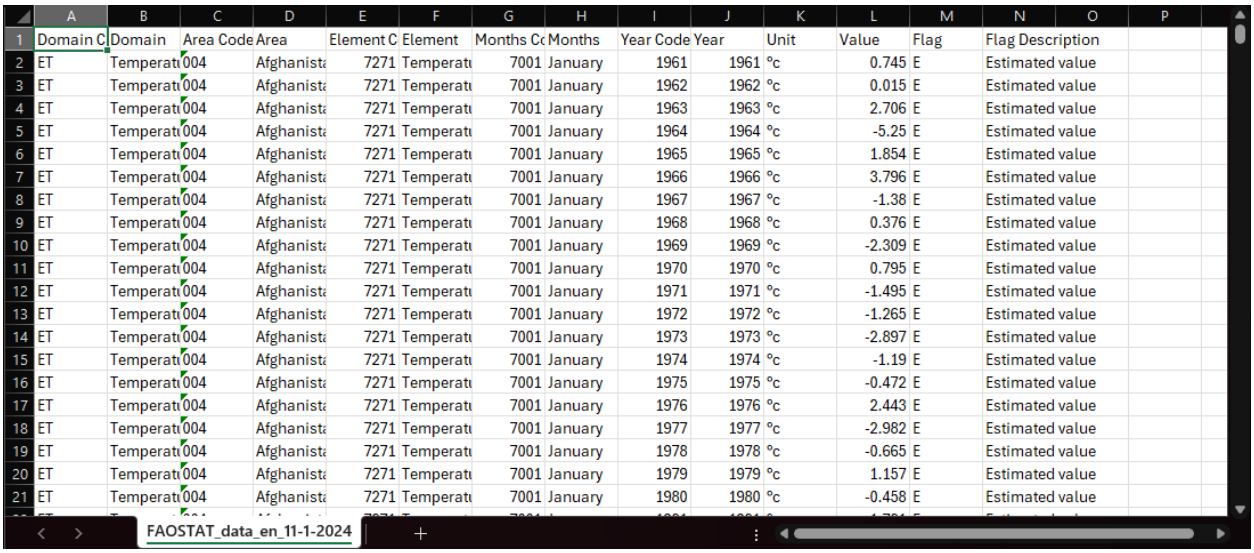
Given that some effects of global warming are already unavoidable, it is crucial to implement adaptation strategies. These include building resilient infrastructure, such as flood defenses and drought-resistant water systems, and strengthening disaster risk management to protect vulnerable populations from extreme weather events. Enhancing resilience is key to reducing the economic and social impacts of climate change.

International cooperation is essential in addressing global warming. Agreements like the Paris Agreement aim to unite countries in efforts to limit global temperature rise to below 2°C, ideally 1.5°C, above pre-industrial levels. Countries are expected to submit national plans for reducing emissions, increasing renewable energy use, and enhancing climate resilience. Additionally, carbon pricing mechanisms such as carbon taxes and cap-and-trade systems are being explored to provide economic incentives for reducing emissions.

The understanding the data and implementing strategies to combat global warming are diverse and require a multifaceted approach involving mitigation, adaptation, and international cooperation. Transitioning to renewable energy, implementing carbon capture technologies, promoting sustainable land use, and adopting climate-resilient practices are vital components of a global response. Governments, businesses, and individuals must work together to meet the challenge of global warming and ensure a sustainable future for the planet. The maps show over the years how temperature is rising and its detrimental for every life form on earth (NASA, 2024).

Screenshot(s) of DataSet:





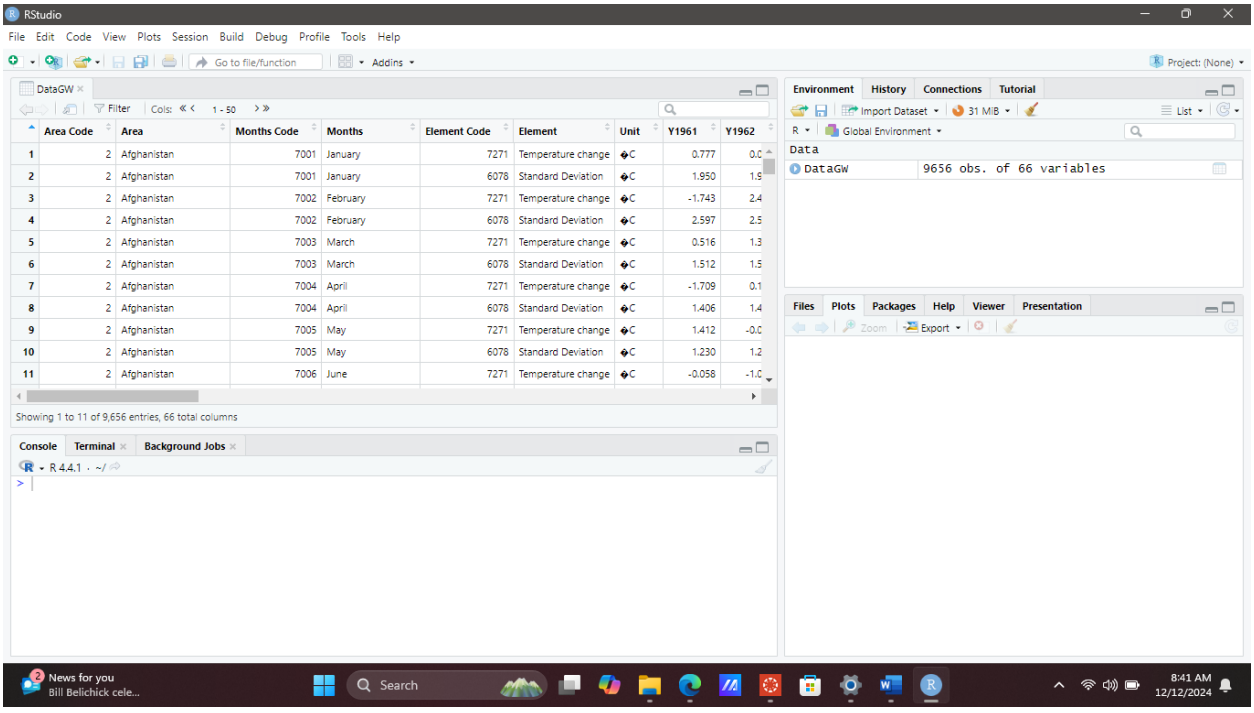
Domain C	Domain	Area Code	Area	Element C	Element	Months Cc	Months	Year Code	Year	Unit	Value	Flag	Flag Description
ET	Temperati	004	Afghanista	7271	Temperatu	7001	January	1961	1961	°c	0.745	E	Estimated value
ET	Temperati	004	Afghanista	7271	Temperatu	7001	January	1962	1962	°c	0.015	E	Estimated value
ET	Temperati	004	Afghanista	7271	Temperatu	7001	January	1963	1963	°c	2.706	E	Estimated value
ET	Temperati	004	Afghanista	7271	Temperatu	7001	January	1964	1964	°c	-5.25	E	Estimated value
ET	Temperati	004	Afghanista	7271	Temperatu	7001	January	1965	1965	°c	1.854	E	Estimated value
ET	Temperati	004	Afghanista	7271	Temperatu	7001	January	1966	1966	°c	3.796	E	Estimated value
ET	Temperati	004	Afghanista	7271	Temperatu	7001	January	1967	1967	°c	-1.38	E	Estimated value
ET	Temperati	004	Afghanista	7271	Temperatu	7001	January	1968	1968	°c	0.376	E	Estimated value
ET	Temperati	004	Afghanista	7271	Temperatu	7001	January	1969	1969	°c	-2.309	E	Estimated value
ET	Temperati	004	Afghanista	7271	Temperatu	7001	January	1970	1970	°c	0.795	E	Estimated value
ET	Temperati	004	Afghanista	7271	Temperatu	7001	January	1971	1971	°c	-1.495	E	Estimated value
ET	Temperati	004	Afghanista	7271	Temperatu	7001	January	1972	1972	°c	-1.265	E	Estimated value
ET	Temperati	004	Afghanista	7271	Temperatu	7001	January	1973	1973	°c	-2.897	E	Estimated value
ET	Temperati	004	Afghanista	7271	Temperatu	7001	January	1974	1974	°c	-1.19	E	Estimated value
ET	Temperati	004	Afghanista	7271	Temperatu	7001	January	1975	1975	°c	-0.472	E	Estimated value
ET	Temperati	004	Afghanista	7271	Temperatu	7001	January	1976	1976	°c	2.443	E	Estimated value
ET	Temperati	004	Afghanista	7271	Temperatu	7001	January	1977	1977	°c	-2.982	E	Estimated value
ET	Temperati	004	Afghanista	7271	Temperatu	7001	January	1978	1978	°c	-0.665	E	Estimated value
ET	Temperati	004	Afghanista	7271	Temperatu	7001	January	1979	1979	°c	1.157	E	Estimated value
ET	Temperati	004	Afghanista	7271	Temperatu	7001	January	1980	1980	°c	-0.458	E	Estimated value

Data Description: (Table)

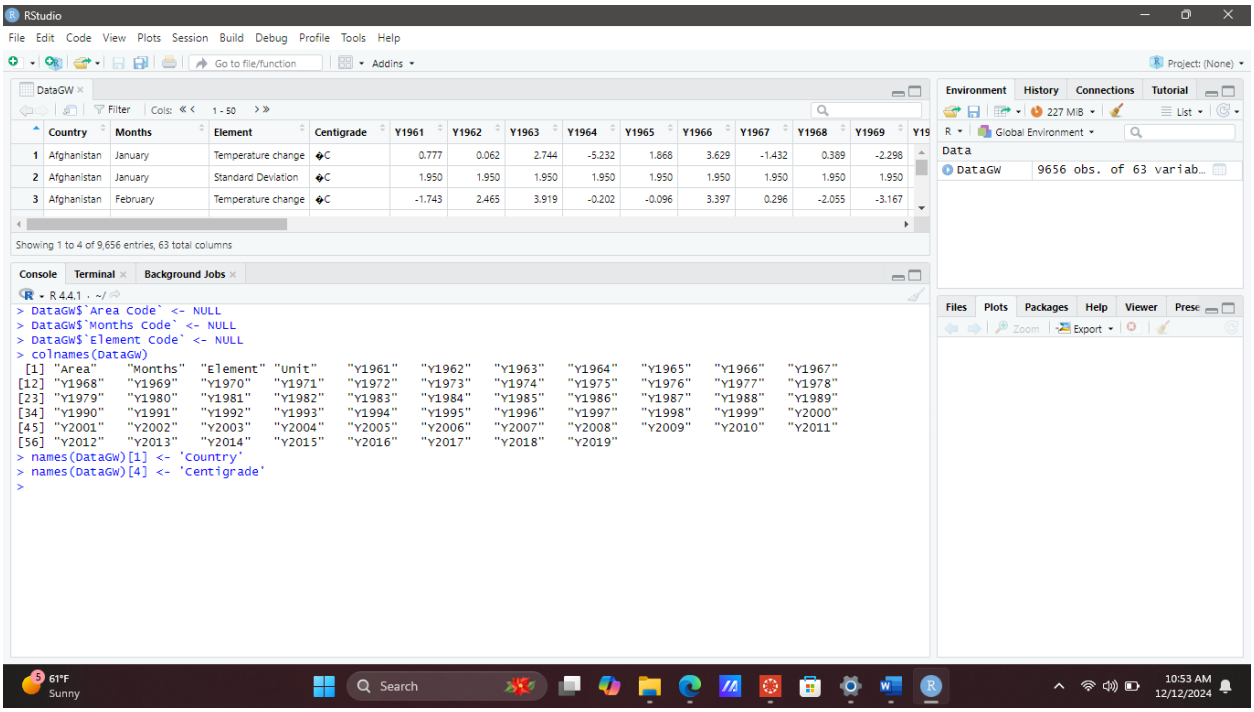
Field Name: Country	Description: The location of where the weather changing.
Field Name: Months	Description: The determination of weather changes within that specific month.
Field Name: Year	Description: The determination of weather changes within that specific year.
Field Name: Unit	Description: The degree of Celsius of weather.
Field Name: Domain	Description: The physical location of weather changes is occurring. For example, “East, West, North, South regions”.
Field Name: Element	Description: Showing that there is a weather change in that region.
Field Name: Year Code	Description: The specific year showing that weather changes occurred.
Field Name: Value	Description: The degree of Celsius of weather pertaining to specific year.

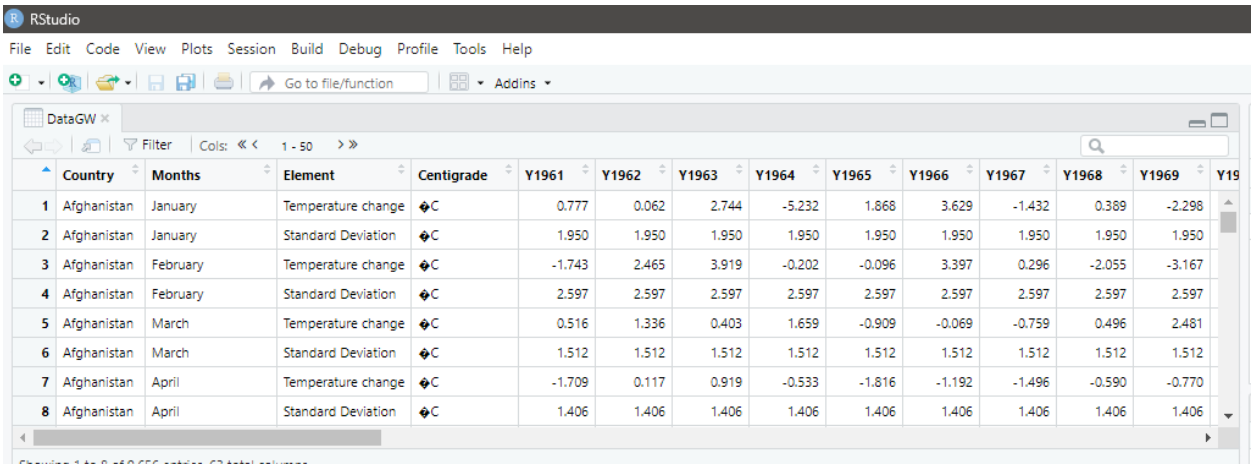
Data Cleaning:

Name Cleaning: **Pre-Cleaning**

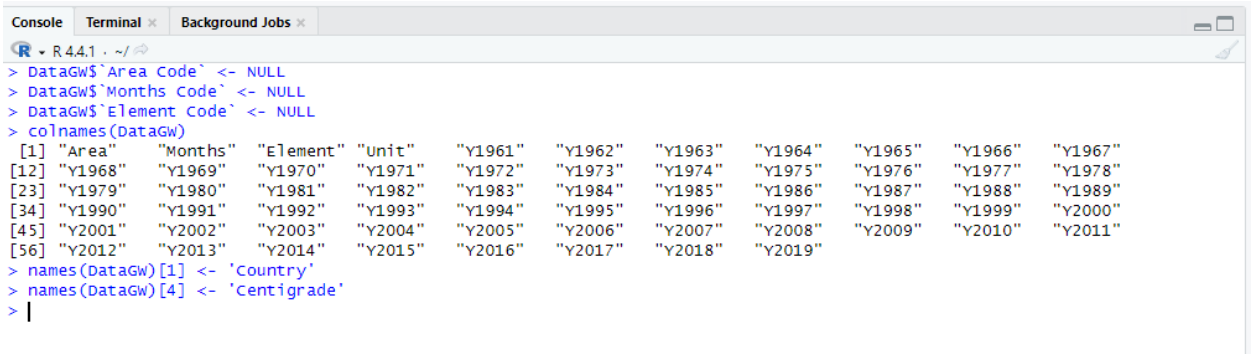


Post-Cleaning:



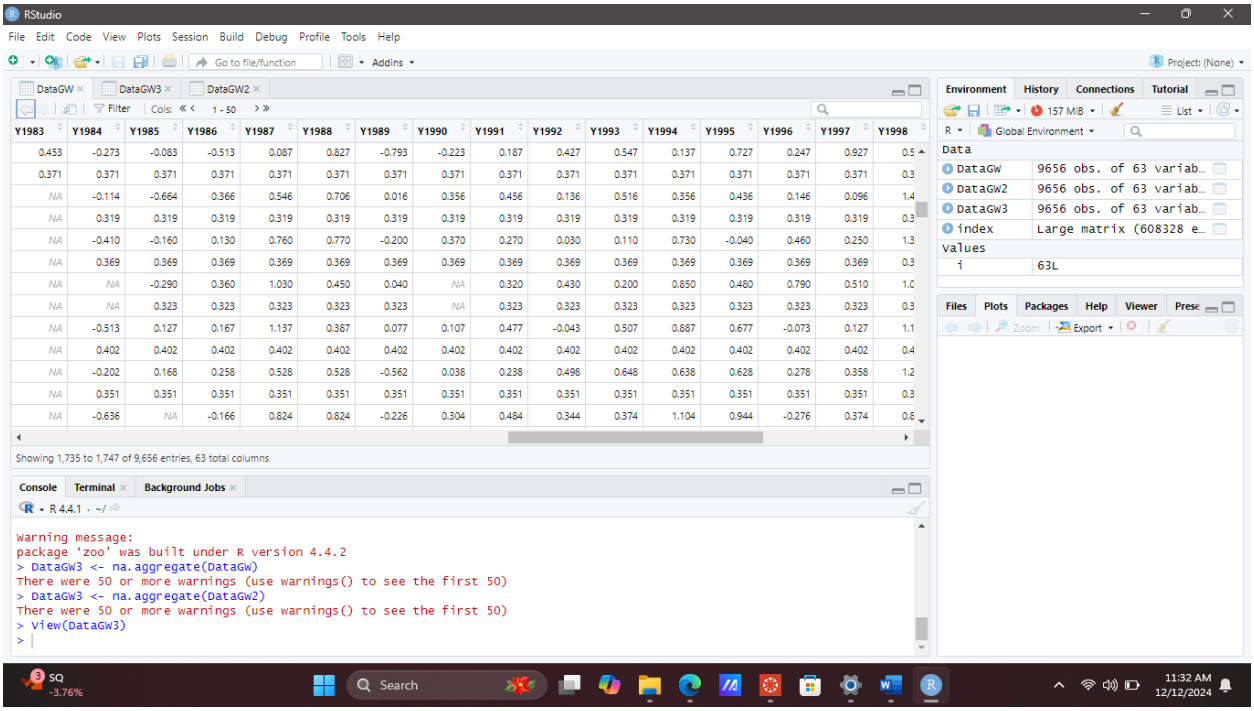


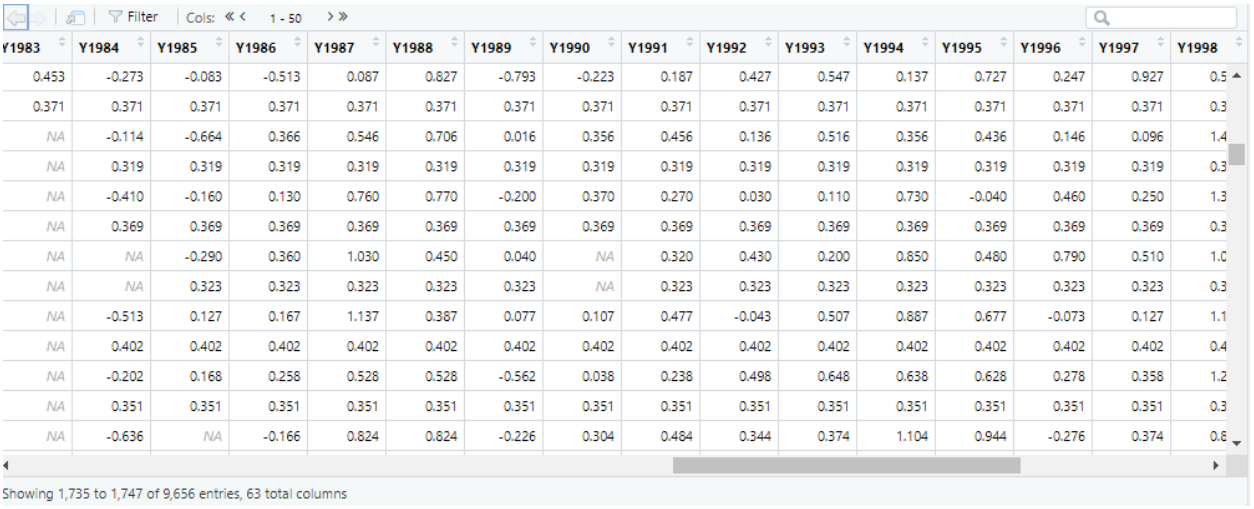
R-Code:



Explanation: From the name cleaning I replace the words, “Area” and “Unit” with the words, “Country” from “Area” and replaced “Unit” with “Centigrade”. As well as I removed invalid information like “Area Code” , “Month Code” and “Element Code”.

Numerical Cleaning: Pre-Cleaning



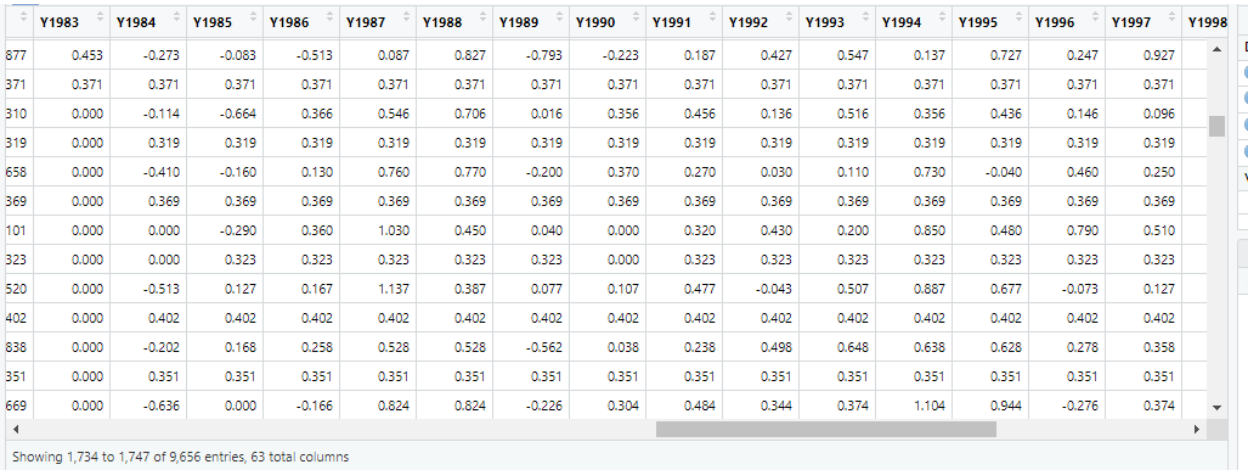


Y1983	Y1984	Y1985	Y1986	Y1987	Y1988	Y1989	Y1990	Y1991	Y1992	Y1993	Y1994	Y1995	Y1996	Y1997	Y1998
0.453	-0.273	-0.083	-0.513	0.087	0.827	-0.793	-0.223	0.187	0.427	0.547	0.137	0.727	0.247	0.927	0.5
0.371	0.371	0.371	0.371	0.371	0.371	0.371	0.371	0.371	0.371	0.371	0.371	0.371	0.371	0.371	0.3
NA	-0.114	-0.664	0.366	0.546	0.706	0.016	0.356	0.456	0.136	0.516	0.356	0.436	0.146	0.096	1.4
NA	0.319	0.319	0.319	0.319	0.319	0.319	0.319	0.319	0.319	0.319	0.319	0.319	0.319	0.319	0.3
NA	-0.410	-0.160	0.130	0.760	0.770	-0.200	0.370	0.270	0.030	0.110	0.730	-0.040	0.460	0.250	1.3
NA	0.369	0.369	0.369	0.369	0.369	0.369	0.369	0.369	0.369	0.369	0.369	0.369	0.369	0.369	0.3
NA	NA	-0.290	0.360	1.030	0.450	0.040	NA	0.320	0.430	0.200	0.850	0.480	0.790	0.510	1.0
NA	NA	0.323	0.323	0.323	0.323	0.323	NA	0.323	0.323	0.323	0.323	0.323	0.323	0.323	0.3
NA	-0.513	0.127	0.167	1.137	0.387	0.077	0.107	0.477	-0.043	0.507	0.887	0.677	-0.073	0.127	1.1
NA	0.402	0.402	0.402	0.402	0.402	0.402	0.402	0.402	0.402	0.402	0.402	0.402	0.402	0.402	0.4
NA	-0.202	0.168	0.258	0.528	0.528	-0.562	0.038	0.238	0.498	0.648	0.638	0.628	0.278	0.358	1.2
NA	0.351	0.351	0.351	0.351	0.351	0.351	0.351	0.351	0.351	0.351	0.351	0.351	0.351	0.351	0.3
NA	-0.636	NA	-0.166	0.824	0.824	-0.226	0.304	0.484	0.344	0.374	1.104	0.944	-0.276	0.374	0.8

Showing 1,735 to 1,747 of 9,656 entries, 63 total columns

Explanation: From the Years 1983, 1985 and 1990 had a few N/A which I used the code to remove the N/A and replace them.

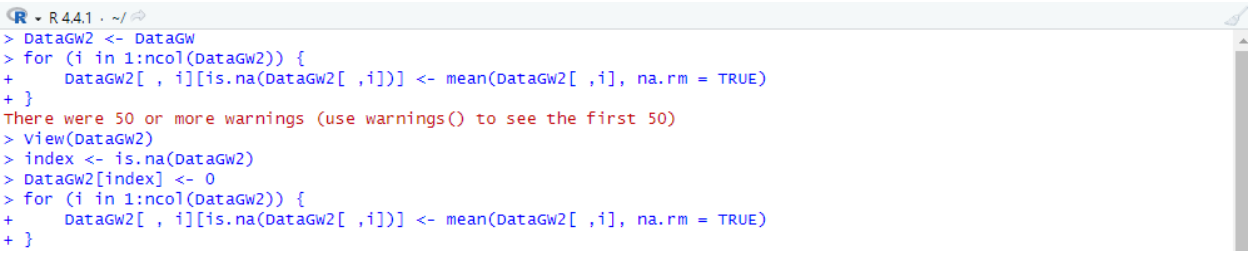
Post-Cleaning:



	Y1983	Y1984	Y1985	Y1986	Y1987	Y1988	Y1989	Y1990	Y1991	Y1992	Y1993	Y1994	Y1995	Y1996	Y1997	Y1998
877	0.453	-0.273	-0.083	-0.513	0.087	0.827	-0.793	-0.223	0.187	0.427	0.547	0.137	0.727	0.247	0.927	
371	0.371	0.371	0.371	0.371	0.371	0.371	0.371	0.371	0.371	0.371	0.371	0.371	0.371	0.371	0.371	
310	0.000	-0.114	-0.664	0.366	0.546	0.706	0.016	0.356	0.456	0.136	0.516	0.356	0.436	0.146	0.096	
319	0.000	0.319	0.319	0.319	0.319	0.319	0.319	0.319	0.319	0.319	0.319	0.319	0.319	0.319	0.319	
658	0.000	-0.410	-0.160	0.130	0.760	0.770	-0.200	0.370	0.270	0.030	0.110	0.730	-0.040	0.460	0.250	
369	0.000	0.369	0.369	0.369	0.369	0.369	0.369	0.369	0.369	0.369	0.369	0.369	0.369	0.369	0.369	
101	0.000	0.000	-0.290	0.360	1.030	0.450	0.040	0.000	0.320	0.430	0.200	0.850	0.480	0.790	0.510	
323	0.000	0.000	0.323	0.323	0.323	0.323	0.323	0.000	0.323	0.323	0.323	0.323	0.323	0.323	0.323	
520	0.000	-0.513	0.127	0.167	1.137	0.387	0.077	0.107	0.477	-0.043	0.507	0.887	0.677	-0.073	0.127	
402	0.000	0.402	0.402	0.402	0.402	0.402	0.402	0.402	0.402	0.402	0.402	0.402	0.402	0.402	0.402	
838	0.000	-0.202	0.168	0.258	0.528	0.528	-0.562	0.038	0.238	0.498	0.648	0.638	0.628	0.278	0.358	
351	0.000	0.351	0.351	0.351	0.351	0.351	0.351	0.351	0.351	0.351	0.351	0.351	0.351	0.351	0.351	
669	0.000	-0.636	0.000	-0.166	0.824	0.824	-0.226	0.304	0.484	0.344	0.374	1.104	0.944	-0.276	0.374	

Showing 1,734 to 1,747 of 9,656 entries, 63 total columns

R-Code:

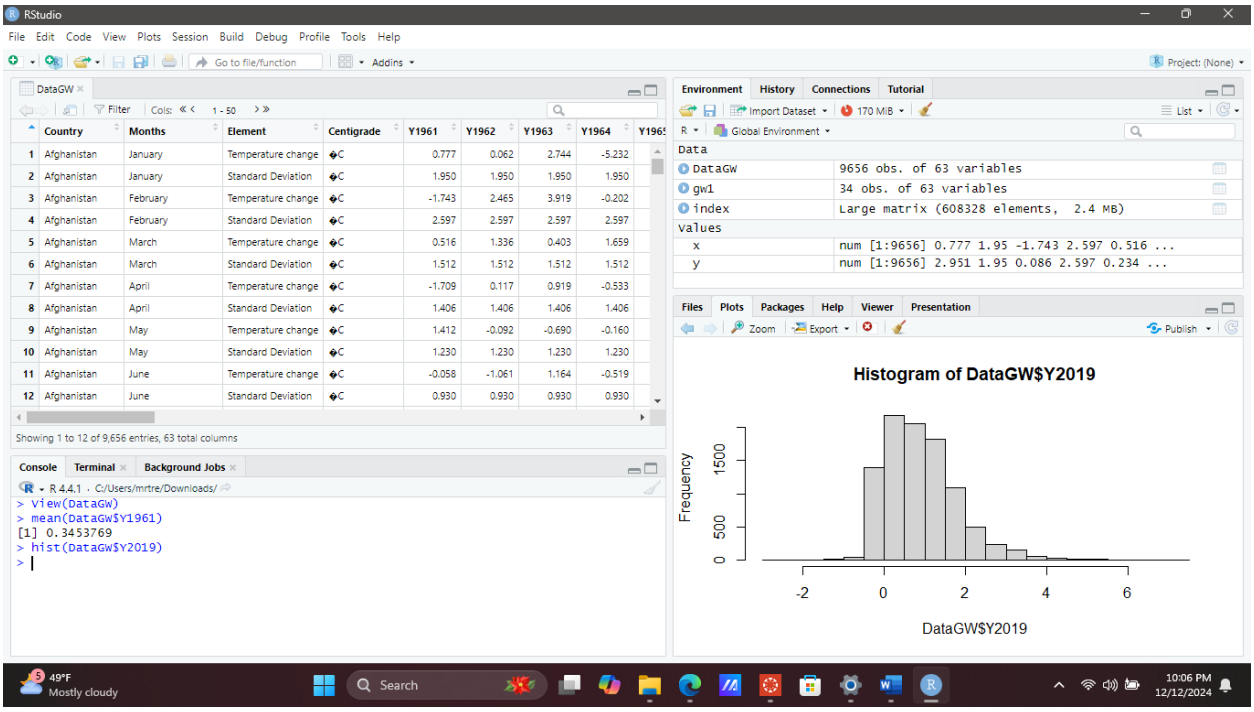


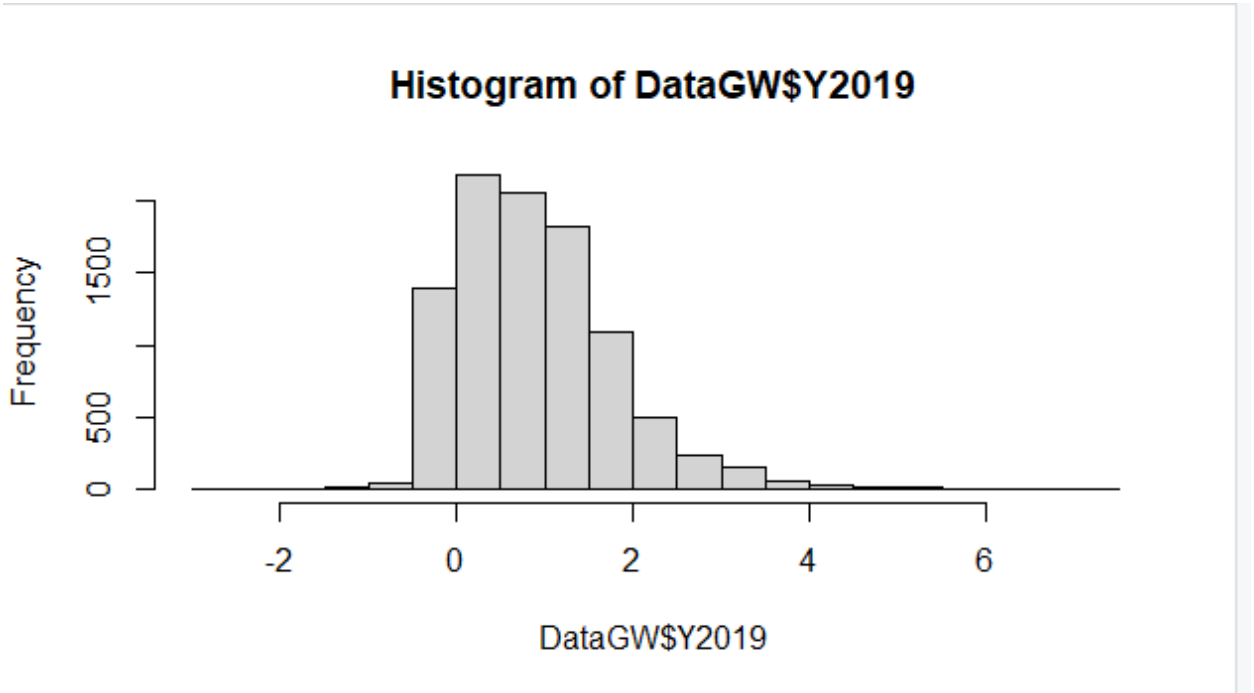
```
R 4.4.1 · ~/
> DataGW2 <- DataGW
> for (i in 1:ncol(DataGW2)) {
+     DataGW2[ , i][is.na(DataGW2[ ,i])] <- mean(DataGW2[ ,i], na.rm = TRUE)
+ }
There were 50 or more warnings (use warnings() to see the first 50)
> view(DataGW2)
> index <- is.na(DataGW2)
> DataGW2[index] <- 0
> for (i in 1:ncol(DataGW2)) {
+     DataGW2[ , i][is.na(DataGW2[ ,i])] <- mean(DataGW2[ ,i], na.rm = TRUE)
+ }
```

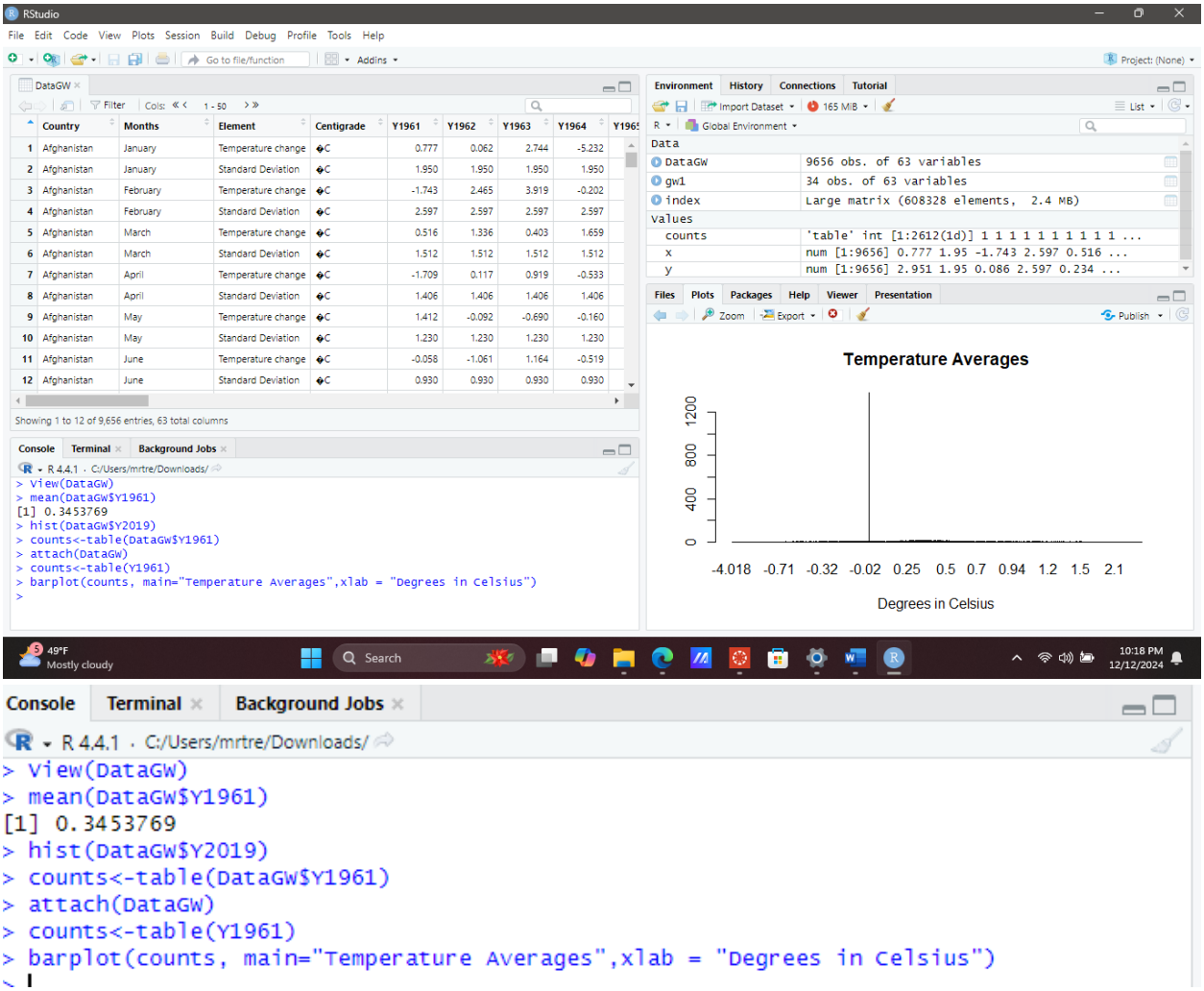
Explanation: I turned all the N/A into zeros, I used code to try to get the mean so I could replace the zeros but I was given a lot of errors with this data set.

Questions of Analysis & Visualizations:

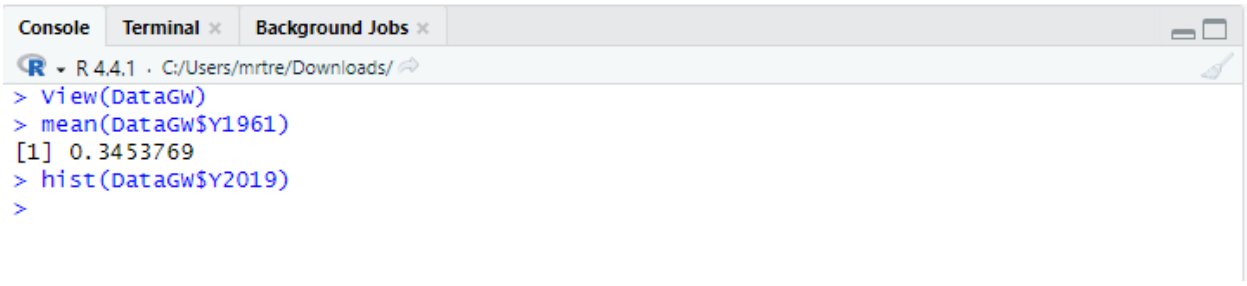
1. What are the temperature patterns in Averages regions in 2019?







(Averages compared from 1961 to 2019.)

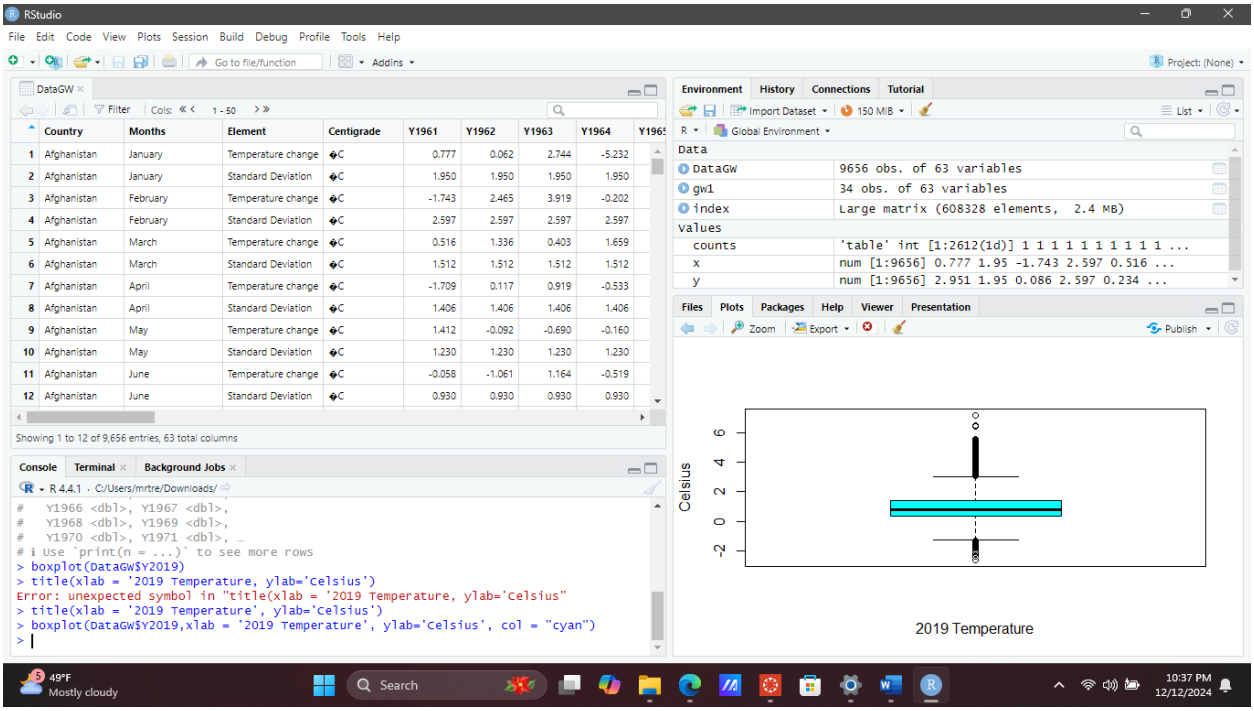


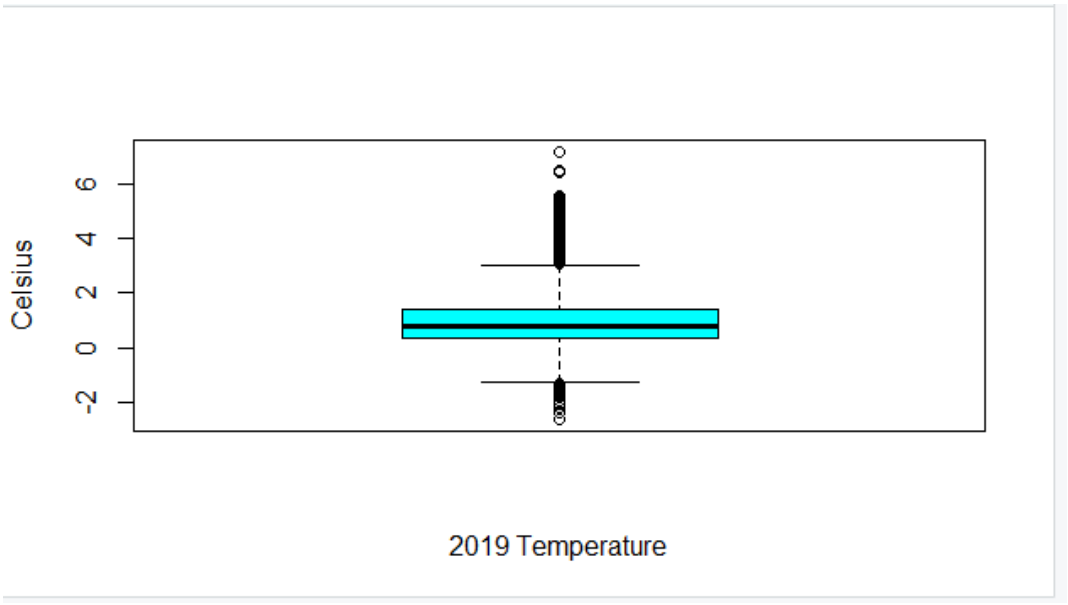
Explanation: The year of 2019 All the average temperature from all countries as the Worlds temperature is around of 1 to 2 Celsius which

for the World that's hot and it's destroying the ecosystem. As one of the Screenshots illustrates the averages from the years 1961 to 2019 which results in average around -0.2 compared to now averages are above and they range between 1 and 2 Celsius. Today, the average Celsius is 1.4 (NASA, 2024).

2. How do United States compare in terms of a specific metric each year, from 1961 to 2019?

3. What is the highest temperature for the specific year of 2019?



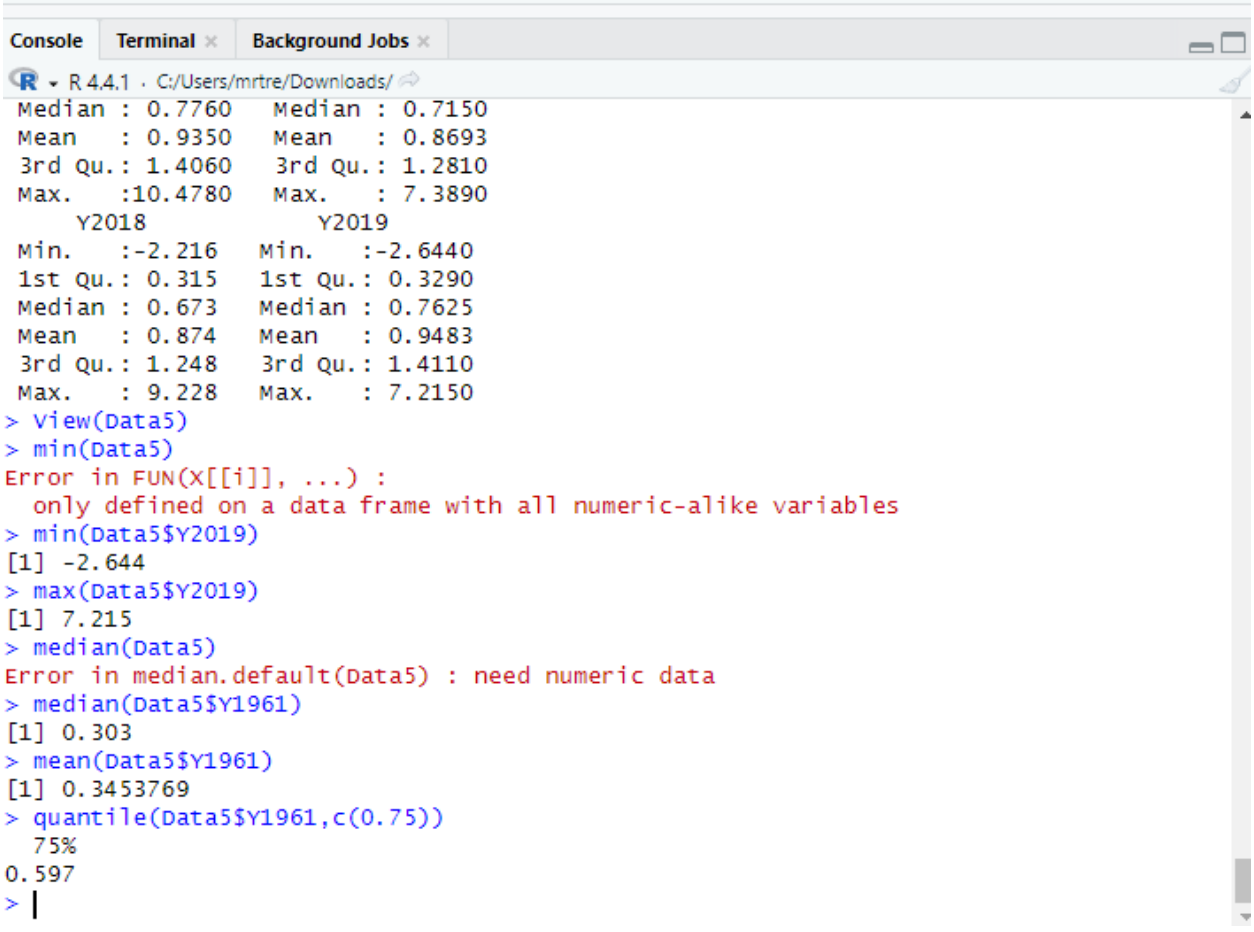


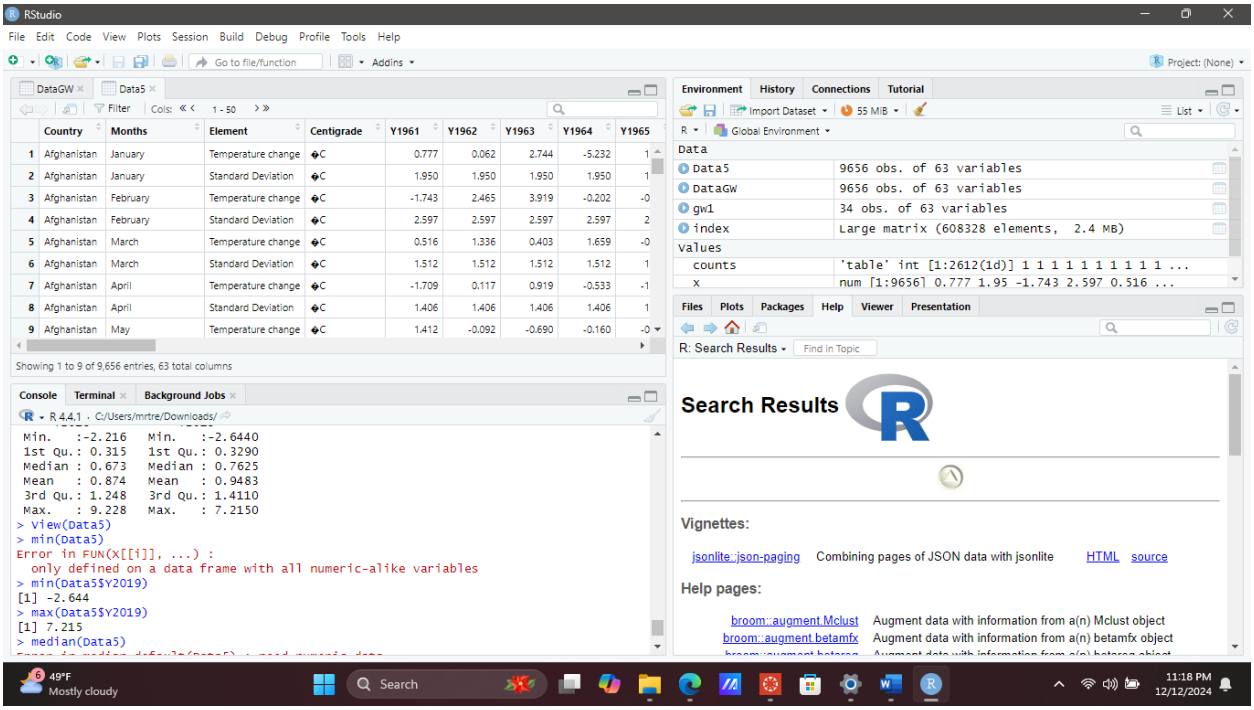

```
Console   Terminal ×   Background Jobs ×
R 4.4.1 · C:/Users/mrtre/Downloads/
> DataGW
# A tibble: 9,656 × 63
   Country   Months Element Centigrade
   <chr>     <chr>  <chr>   <chr>
 1 Afghanis… Janua… Temper… "\xb0C"
 2 Afghanis… Janua… Standa… "\xb0C"
 3 Afghanis… Febru… Temper… "\xb0C"
 4 Afghanis… Febru… Standa… "\xb0C"
 5 Afghanis… March  Temper… "\xb0C"
 6 Afghanis… March  Standa… "\xb0C"
 7 Afghanis… April  Temper… "\xb0C"
 8 Afghanis… April  Standa… "\xb0C"
 9 Afghanis… May    Temper… "\xb0C"
10 Afghanis… May    Standa… "\xb0C"
# i 9,646 more rows
# i 59 more variables: Y1961 <dbl>,
#   Y1962 <dbl>, Y1963 <dbl>,
#   Y1964 <dbl>, Y1965 <dbl>,
#   Y1966 <dbl>, Y1967 <dbl>,
#   Y1968 <dbl>, Y1969 <dbl>,
#   Y1970 <dbl>, Y1971 <dbl>, …
# i Use `print(n = ...)` to see more rows
> boxplot(DataGW$Y2019)
> title(xlab = '2019 Temperature, ylab='Celsius')
Error: unexpected symbol in "title(xlab = '2019 Temperature, ylab='Celsius"
> title(xlab = '2019 Temperature', ylab='Celsius')
> boxplot(DataGW$Y2019,xlab = '2019 Temperature', ylab='Celsius', col = "cyan")
>
```

Explanation: The box plot chart shows in 2019 that in the countries Celsius is about 1C higher or less. According to the article Worlds average of Celsius is around 1, 2023 was the warmest year since global records began in 1850 by a

wide margin It was 2.12 °F (1.18 °C) above the 20th-century average (Dahlman,2024).

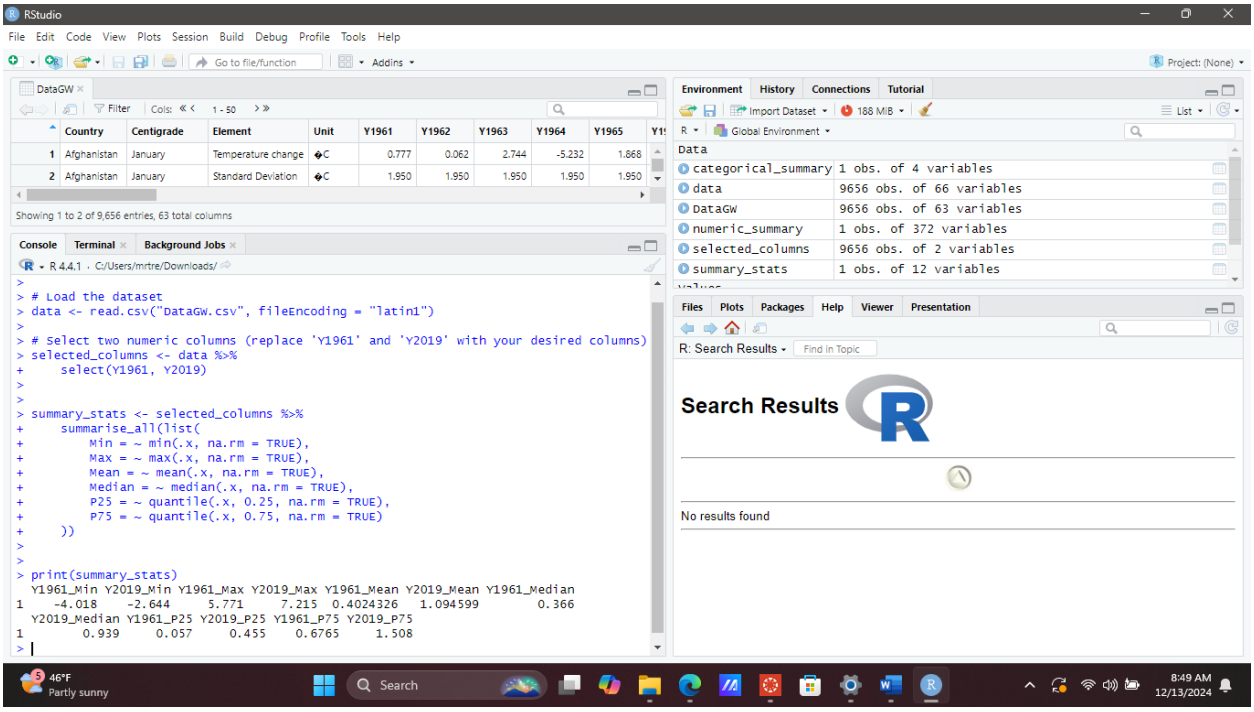
Statistical Summary and Script: Min,Max,Mean,Median,Percentile





Explanation: 75% of the averages of temperature in the year 1961 is about 0.597 round it off its about 0.6. Compared to the average temperature to 2019, which is about 1.1, so the level of heat has risen about 0.5. It might not sound like a big percentile but its affect economies around the world.

Statistical Summary:

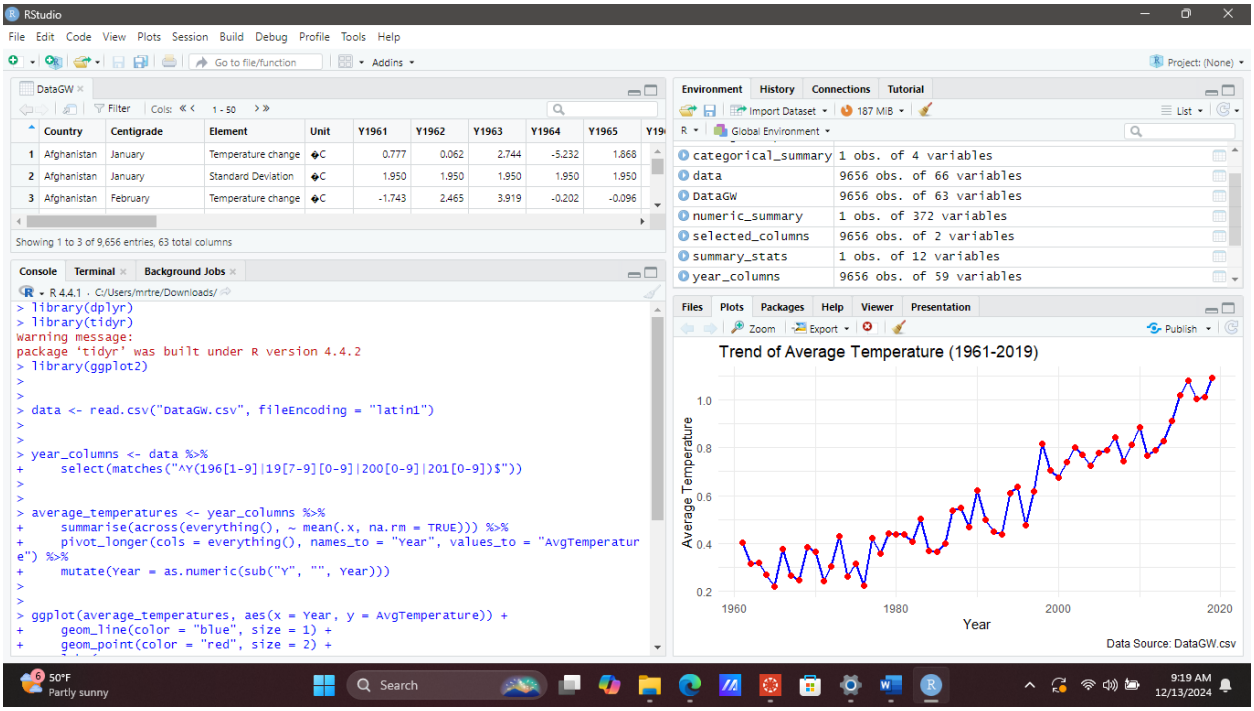


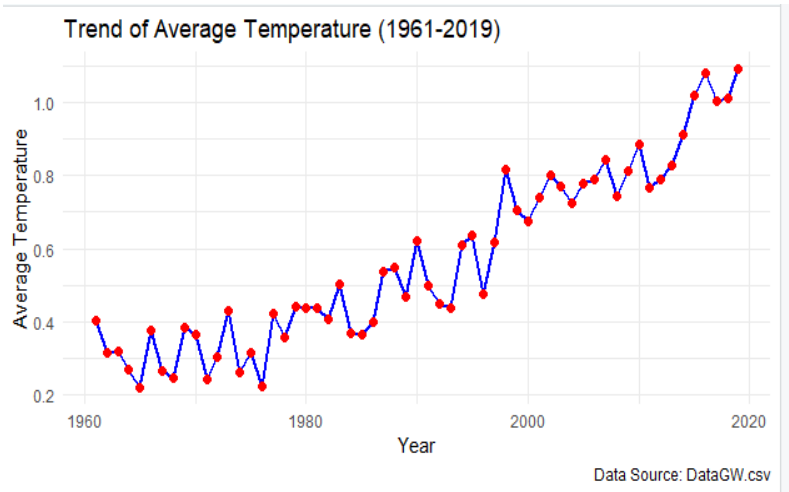
R- Code:

```
Console  Terminal  Background Jobs
R 4.4.1 · C:/Users/mrtre/Downloads/
> library(dplyr)
>
> # Load the dataset
> data <- read.csv("DataGW.csv", fileEncoding = "latin1")
>
> # Select two numeric columns (replace 'Y1961' and 'Y2019' with your desired columns)
> selected_columns <- data %>%
+     select(Y1961, Y2019)
>
>
> summary_stats <- selected_columns %>%
+     summarise_all(list(
+         Min = ~ min(.x, na.rm = TRUE),
+         Max = ~ max(.x, na.rm = TRUE),
+         Mean = ~ mean(.x, na.rm = TRUE),
+         Median = ~ median(.x, na.rm = TRUE),
+         P25 = ~ quantile(.x, 0.25, na.rm = TRUE),
+         P75 = ~ quantile(.x, 0.75, na.rm = TRUE)
+     ))
>
>
> print(summary_stats)
  Y1961_Min Y2019_Min Y1961_Max Y2019_Max Y1961_Mean Y2019_Mean Y1961_Median
1    -4.018    -2.644     5.771     7.215  0.4024326  1.094599        0.366
  Y2019_Median Y1961_P25 Y2019_P25 Y1961_P75 Y2019_P75
1        0.939     0.057     0.455    0.6765     1.508
>
```

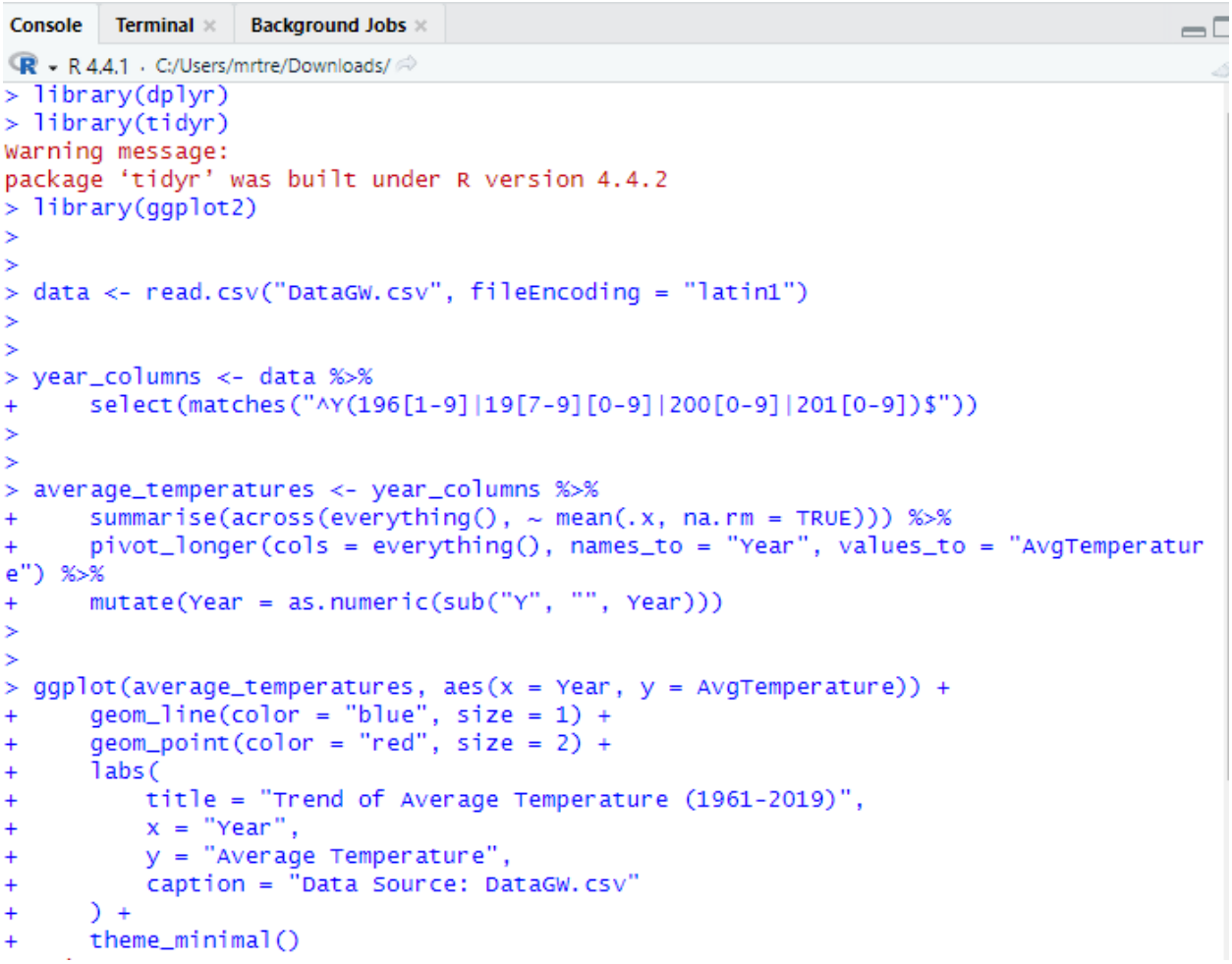
Explanation: In the statistical summary from the year 1961 to the year 2019, the min, max, mean, median show a drastic change in the weather as the numbers are elevated even with then min that contains a negative integer as for 1961 was -4.018 compared to 2019 which is -2.644. As in integers the closer you are to zero shows an increase. So, overall, the data statistics show that from the year 1961 to 2019, the world is getting warmer, which is leading to global warming and the destruction of the ecosystem.

R-Script:





R-Code:



Explanation: The written code compares the data given in the data set to show the trends of the weather patterns throughout the years since 1961, as the graph illustrates over the years the temperature is getting warmer which is bad for the ecosystem.

Works Cited

"Global Surface Temperature." *NASA*, NASA, 7 Feb. 2024, climate.nasa.gov/vital-signs/global-temperature/?intent=121.

"World of Change: Global Temperatures." *NASA*, NASA, earthobservatory.nasa.gov/world-of-change/global-temperatures. Accessed 12 Dec. 2024.

Dahlman, Rebecca Lindsey AND LuAnn. "Climate Change: Global Temperature." *NOAA Climate.Gov*, 18 Jan. 2024, www.climate.gov/news-features/understanding-climate/climate-change-global-temperature.

Dewan, Angela. "The World Just Marked a Year above a Critical Climate Limit Scientists Have Warned About." *CNN*, Cable News Network, 8 Feb. 2024, www.cnn.com/2024/02/08/climate/global-warming-limit-climate-intl/index.html.